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C03/AT2

1) Shepherd Boy case study - TP, FP, FN and TN

Ans

True Positive (TP)

A True Positive occurs when the wolf is actually present and the shepherd boy correctly gives the warning

Example: A real wolf approaches the flock and the boy cries "Wolf!". The danger is correctly identified

TP = Wolf present + Correct warning

False Positive (FP)

A false positive occurs when there is no wolf, but the boy gives a warning. This happens when he cries "Wolf!" as a joke even though there is no wolf.

FP = No Wolf + False warning

This is called a false alarm.

False Negative (FN)

A False Negative occurs when a real wolf is present but the danger is not effectively recognized. In the story, the boy sees the real wolf and calls for help.

FN = Wolf Present + Danger not effectively recognized

True Negative (TN)

A True Negative occurs when there is no wolf and no unnecessary warning is given

TN = No wolf + No warning

classification of Result

Actual Condition	warning Given	Result
wolf Present	Yes	TP
wolf absent	Yes	FP
wolf Present	No effectively missed	FN
wolf absent	No warning	TN

Q) Cat image classification

Ans The dataset contains 100 images among them 30 images contain cats and 70 images do not contain cats.

Step 1: Calculate TP, FP, FN and TN

True Positive

The model correctly detects 25 cat images

$$\boxed{TP = 25}$$

False Negative

There are 30 actual cat images, but only 25 are detected

$$FN = 30 - 25 = 5$$

$$\boxed{FN = 5}$$

True Negative

The model correctly identifies 50 images that do not contain cats:

$$\boxed{TN = 50}$$

False Positive

There are 70 non-cat images and 50 are correctly

$$FP = 70 - 50 = 20$$

$$\boxed{FP = 20}$$

Result
TP
FP

Confusion Matrix

	Actual cat	Actual No cat
Predicted cat	TP = 25	FP = 20
Predicted No cat	FN = 5	TN = 50

Precision

Precision tells us how many images predicted as cats are actually cats.

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{25}{25 + 20} = \frac{25}{45} = 55.6\%$$

Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$= \frac{25}{25 + 5} = \frac{25}{30} = 83.3\%$$

F1 Score

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$F1 \approx 0.667$$

$$\boxed{F1 = 66.7\%}$$

Final Result

$$TP - 25$$

$$FP - 20$$

$$FN - 5$$

$$TN - 50$$

$$\text{Precision} - 55.6\%$$

$$\text{Recall} - 83.3\%$$

$$F1 \text{ score} - 66.7\%$$

3) Email spam classification using Naive Bayes

Ans

Given Dataset

offer	Free	length	Class
Yes	Yes	short	Spam
Yes	No	long	Not spam
No	Yes	short	Spam
No	No	long	Not spam
Yes	Yes	Long	Spam
NO	NO	short	NOT spam

1) Prior Probabilities

There are 6 emails in total

* spam = 3

* Not spam = 3

Therefore .

$$P(\text{spam}) = \frac{3}{6} = 0.5$$

$$P(\text{Not spam}) = \frac{3}{6} = 0.5$$

2) Conditional Probabilities

For spam

$$P(\text{offer} = \text{Yes} / \text{spam}) = \frac{2}{3}$$

$$P(\text{Free} = \text{No} / \text{spam}) = \frac{0}{3} = 0$$

$$P(\text{Length} = \text{short} / \text{spam}) = \frac{2}{3}$$

For Not spam

$$P(\text{offer} = \text{Yes} / \text{Not spam}) = \frac{1}{3}$$

$$P(\text{Free} = \text{No} / \text{Not spam}) = \frac{2}{3}$$

$$P(\text{Length} = \text{short} / \text{Not spam}) = \frac{1}{3}$$

classify, the New Email

The test email has

* Offer = Yes

* Free = NO

* Length = short

Probability for spam

$$P(\text{spam} | x) \propto 0.5 \times \frac{2}{3} \times 0 \times \frac{2}{3} \\ = 0$$

Probability for Not Spam

$$P(\text{Notspam} | x) \propto 0.5 \times \frac{1}{3} \times \frac{2}{3} \times \frac{1}{3} \\ = \frac{1}{27}$$

Therefore

Prediction = Not spam

4) comparison Using WEKA

In this for comparison using the WEKA Naive Bayes

In WEKA:

- 1) Load the email dataset
- 2) Select Classifier
- 3) Select Bayes
- 4) Select Naive Bayes
- 5) Run the classifier
- 6) Test the new email with
 - * Offer = yes
 - * Free = No
 - * Length = short

The expected classification from the given Probability
NOT SPAM

The exact probability values displayed by WEKA

Assad
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